

Use the equation $y = -\frac{1}{2}x - 3$ for questions 1 and 2.

1. What is the form of this linear equation?

This equation is in slope-intercept form.

2. Rewrite this equation in standard form.

$$y = -\frac{1}{2}x - 3$$

$$+\frac{1}{2}x \quad +\frac{1}{2}x$$

$\frac{1}{2}x + y = -3$ or any
equivalent equation in
standard form.

Use the equation $y - 7 = \frac{2}{3}(x - 9)$ for questions 3 and 4.

3. What is the form of this linear equation?

This equation is in point-slope form.

4. Rewrite this equation in standard form.

$$\begin{aligned} y - 7 &= \frac{2}{3}(x - 9) \\ y - 7 &= \frac{2}{3}x - \frac{2}{3} \cdot \frac{9}{1} \\ y - 7 &= \frac{2}{3}x - 6 \end{aligned}$$

$-\frac{2}{3}x + y = 1$ or any
equivalent equation in
standard form.

Use the equation $3x + 5y = -2$ for questions 5 and 6.

5. What is the form of this linear equation?

This equation is in standard form.

6. Rewrite this equation in slope-intercept form.

$$\begin{array}{rcl} 3x + 5y = -2 & \frac{5y}{5} = \frac{-3x}{5} - \frac{2}{5} \\ \underline{-3x} & \underline{-3x} & \end{array}$$

$$5y = -3x - 2 \quad \boxed{y = \frac{-3}{5}x - \frac{2}{5}}$$

Use the equation $y + 1 = 2(x + 4)$ for questions 7 through 10.

$$y - y_1 = m(x - x_1)$$

7. What is the slope of the line?

$$\begin{array}{rcl} y + 1 = 2x + 8 \\ y = 2x + 7 \\ y = mx + b \\ m = 2 \quad b = 7 \end{array}$$

The slope is 2.

8. Rewrite this equation in slope-intercept form.

$$\boxed{y = 2x + 7}$$

9. Identify the y -intercept of the line.

The y -intercept is (0,7)

10. Rewrite the equation in standard form.

$$\begin{array}{rcl} y + 1 = 2x + 8 & y = 2x + 7 \\ \underline{-1} & \underline{-1} & \underline{-2x} \quad \underline{-2x} \end{array}$$

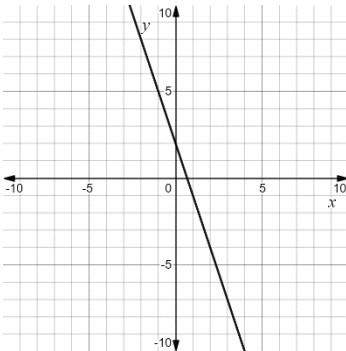
$-2x + y = 7$ or any equivalent equation in standard form.

Algebra 1
Unit 3
Lesson 2 Homework

Name Answer Key
Date _____
Period _____

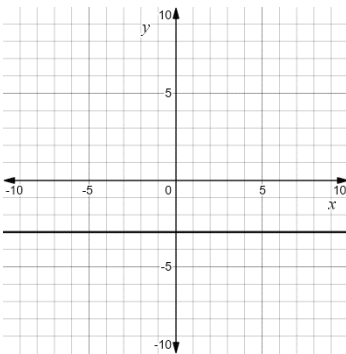
For questions 1 through 4, determine the slope of the line shown.

1.



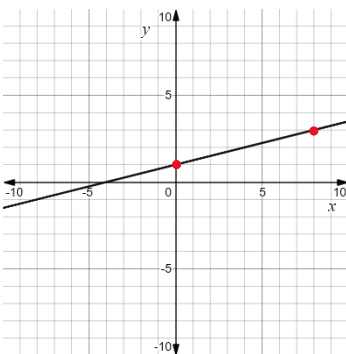
$$m = \underline{\quad -3 \quad}$$

2.



$$m = \underline{\quad 0 \quad}$$

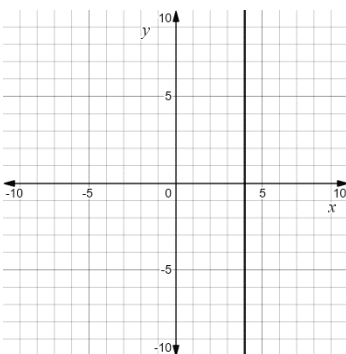
3.



$$\begin{aligned} (x_1, y_1) &= (0, 1) \\ (x_2, y_2) &= (8, 3) \\ m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{3 - 1}{8 - 0} = \frac{2}{8} \\ &= \frac{1}{4} \end{aligned}$$

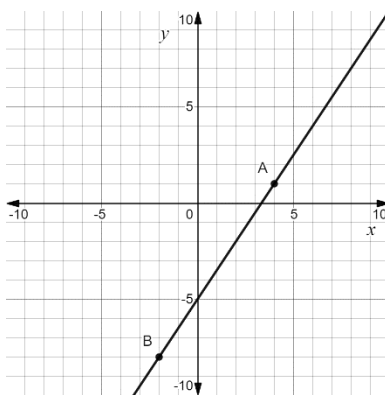
$$m = \underline{\quad \frac{1}{4} \quad}$$

4.



$$m = \underline{\quad \text{undefined} \quad}$$

For questions 5 through 8, the graph of a line with points A and B labeled is shown.



5. Write the equation of the line in point-slope form using point A .

$$m = \frac{3}{2} \quad (4, 1) \quad y - 1 = \frac{3}{2}(x - 4)$$

6. Write the equation of the line in point-slope form using point B .

$$m = \frac{3}{2} \quad (-2, -8) \quad y + 8 = \frac{3}{2}(x + 2)$$

7. Rewrite the equation in point-slope form from point A in slope-intercept form.

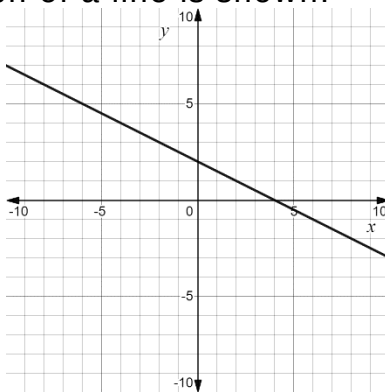
$$y - 1 = \frac{3}{2}x - \frac{12}{2} \rightarrow y - 1 + 1 = \frac{3}{2}x - 6 + 1 \quad y = \frac{3}{2}x - 5$$

8. Rewrite the equation in point-slope form from point B in standard form.

$$y + 8 = \frac{3}{2}x + 3 \rightarrow y - \frac{3}{2}x = 3 - 8$$

$$-\frac{3}{2}x + y = -5 \text{ or } -3x + 2y = -10$$

For questions 9 and 10, the graph of a line is shown.



9. Write the equation of the line in point-slope form.

$$m = -\frac{1}{2} \quad (0, 2) \quad y - 2 = -\frac{1}{2}(x - 0)$$

10. Rewrite the equation from question 9 in standard form.

$$y - 2 = -\frac{1}{2}x \rightarrow y - 2 + 2 = -\frac{1}{2}x + 2 \rightarrow y + \frac{1}{2}x = -\frac{1}{2}x + 2 + \frac{1}{2}x$$

$$2(y + \frac{1}{2}x = 2) \rightarrow x + 2y = 4$$

For questions 1 through 4, a table of values representing a linear relationship is shown.

x	-1	0	3	8
y	-3.4	-3	-1.8	0.2

1. Find the slope of the line. Show your work.

$$(x_1, y_1): (0, -3) \quad \frac{0.2 + 3}{8 - 0} = \frac{3.2}{8} = 0.4$$

$$(x_2, y_2): (8, 0.2)$$

2. Write the equation of the line modeling this table in point-slope form.

$$y + 3 = 0.4(x - 0)$$

3. Rewrite the equation of the line in slope-intercept form.

$$y + 3 = 0.4x$$

$$-3 \quad -3$$

$$y = 0.4x - 3$$

4. Rewrite the equation of the line in standard form.

$$-\frac{2}{5}x + y = -3$$

$$2x - 5y = 15$$

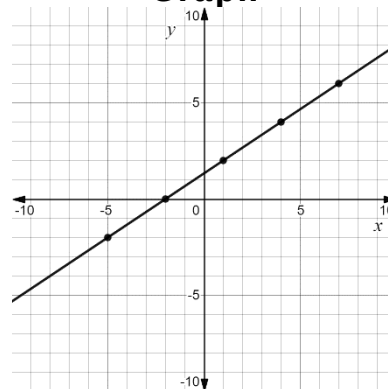
$$5\left(\frac{2}{5}x - y = 3\right)$$

A table and graph for a linear function, $g(x)$ are shown. Use the table and graph for questions 5 and 6.

Table

x	$g(x)$
-5	-2
1	2
7	6

Graph



5. Write the equation of the line in point-slope form.

$$y - 2 = \frac{2}{3}(x - 1)$$

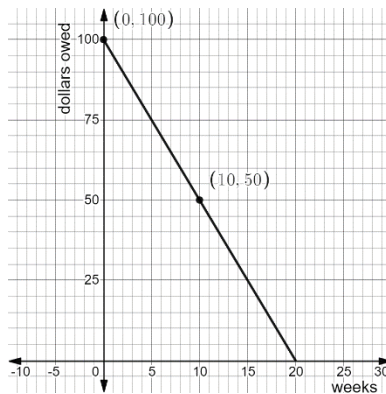
6. Write the equation of the line in slope-intercept form.

$$y - 2 = \frac{2}{3}x - \frac{2}{3}$$

$$+2 \quad +2$$

$$y = \frac{2}{3}x + \frac{4}{3}$$

Brandon borrowed \$100 from his grandmother for summer camp. He agrees to repay her \$5 per week until he has paid the money back. The graph of this linear relationship is shown. Use the graph to answer questions 7 through 10.



7. Find the slope of the line.

$$-5$$

8. Write the equation of the line in point slope form.

$$y - 100 = -5(x - 0)$$

9. Write the equation of the line in slope-intercept form.

$$y - 100 = -5x$$

$$+100 \quad +100$$

$$y = -5x + 100$$

10. Write the equation of the line in standard form.

$$5x + y = 100$$

Algebra 1
Unit 3
Lesson 4 Homework

Name Answer Key
Date _____
Period _____

A line passes through the points $(-3, 7)$ and $(4, -2)$. Use this information for questions 1 through 3.

(x_1, y_1) (x_2, y_2)

1. Find the slope of the line. Show your work.

$$\frac{-2 - 7}{4 - (-3)} = \frac{-9}{7}$$

2. Write the equation of the line in point-slope form.

$$y - 7 = \frac{-9}{7}(x + 3)$$

3. Rewrite the equation of the line in standard form.

$$y - 7 = \frac{-9}{7}x - \frac{27}{7}$$

$$y = \frac{-9}{7}x + \frac{22}{7}$$

$$7\left(\frac{9}{7}x + y = \frac{22}{7}\right)$$

$$9x + 7y = 22$$

Lee uses reclaimed rainwater in his garden. At 2 p.m., the water level in his rain barrel was 2.4 inches. During an afternoon rain shower, from 2 p.m. to 5 p.m., the rain fell at a steady rate. At 5 p.m., the water level in his rain barrel was 3.95. Use this information for questions 4 through 7.

4. Determine two ordered pairs on the graph of this relationship.

$(2, 2.4)$ $(5, 3.95)$

5. Use the two ordered pairs to find the slope of the line.

$$\frac{3.95 - 2.4}{5 - 2} = \frac{1.55}{3} = \frac{31}{60}$$

6. Use an ordered pair and the slope to write the equation of the line in point-slope form.

$$y - 2.4 = \frac{31}{60}(x - 2)$$

7. Use the two ordered pairs to write the equation the equation of the line in slope-intercept form.

$$y - 2.4 = \frac{31}{60}x - \frac{62}{60}$$

$$+2.4 \qquad +2.4$$

$$y = \frac{31}{60}x + \frac{41}{30}$$

A line has an x -intercept of -5 and a y -intercept of 3 . Use this information for questions 8 through 10.

8. Find the slope of the line.

$$(x_1, y_1): (-5, 0)$$

$$(x_2, y_2): (0, 3)$$

$$\frac{3 - 0}{0 - (-5)} = \frac{3}{5}$$

9. Write the equation of the line in slope-intercept form.

$$y - 0 = \frac{3}{5}(x + 5)$$

$$y = \frac{3}{5}x + 3$$

10. Write the equation of the line in standard form.

$$-\frac{3}{5}x + y = 3$$

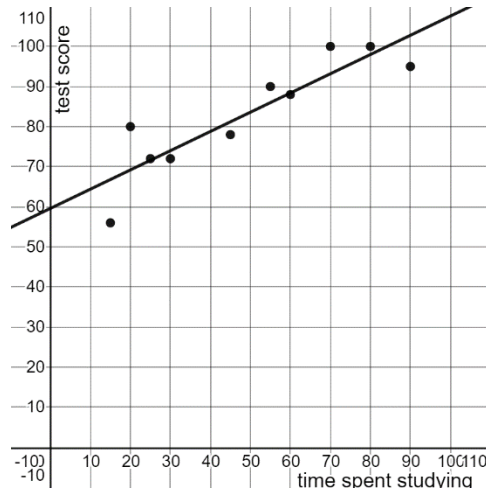
$$5\left(\frac{3}{5}x - y = -3\right)$$

$$3x - 5y = -15$$

Algebra 1
Unit 3
Lesson 5 Homework

Name Answer Key
Date _____
Period _____

For questions 1 through 5, the scatter plot shows the relationship between the time students spent studying, in minutes, for a test and the test score earned.



The equation $y = 0.48x + 59.58$ can be used to model the relationship between the time spent studying in minutes, x , and the test score earned, y .

1. Find the slope of the model.

0.48

2. Determine the y -intercept of the model.

59.58

3. What does the slope represent in the context?

The test score will increase by 0.48 points per minute of study.

4. Determine an appropriate domain in the context.

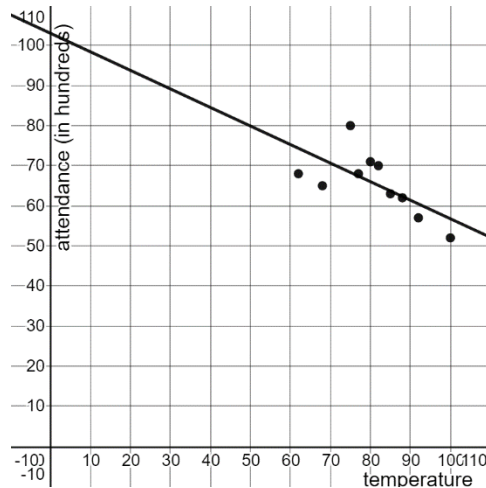
$\{x | 0 \leq x \leq 84.2\}$

$$\begin{aligned} 100 &= 0.48x + 59.58 \\ -59.58 &\quad -59.58 \\ \hline 40.42 &= \frac{0.48x}{0.48} \\ 84.2 &= x \end{aligned}$$

5. Determine an appropriate range in the context.

$\{y | 59.58 \leq y \leq 100\}$

For questions 6 through 10, the scatter plot shows the relationship between the average daily temperature and the daily attendance at a theme park.



The equation $y = -0.46x + 103.04$ can be used to model the relationship between the temperature, x , and the attendance at a theme park, y .

6. Find the slope of the model. Show your work.

-0.46

7. Determine the y -intercept of the model.

103.04

8. What does the slope represent in the context?

Attendance will drop by 0.46 hundred people per degree of temperature increase.

9. Determine an appropriate domain in the context.

Answers may vary.

Sample answer: $\{x | 50 \leq x \leq 100\}$

10. Determine an appropriate range in the context.

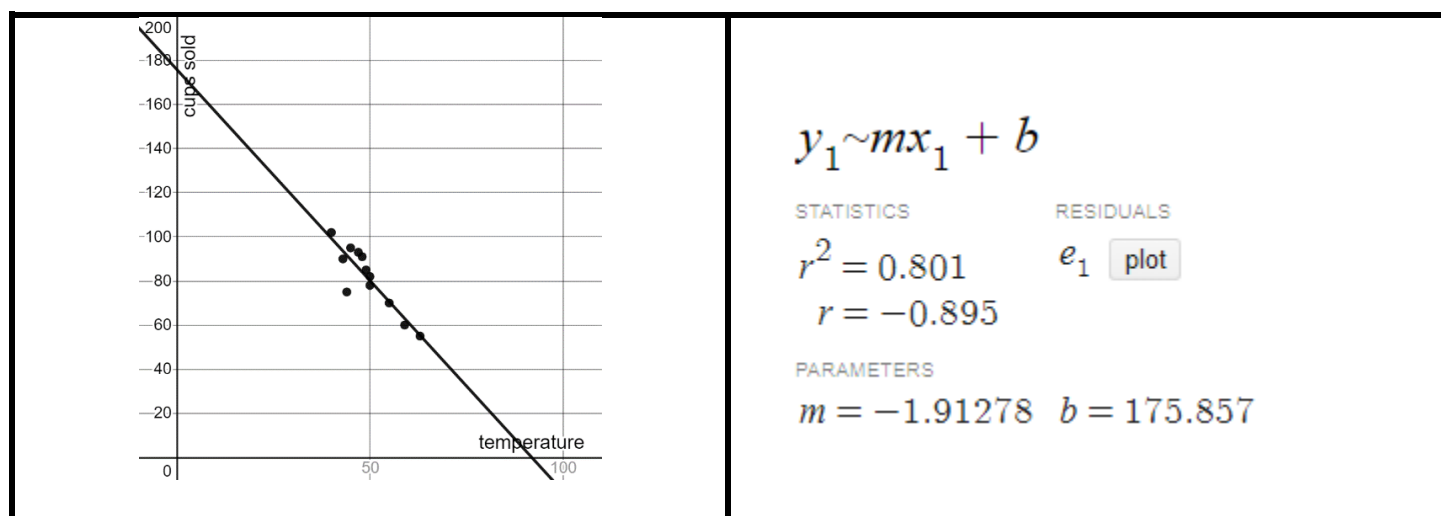
$\{y | y \geq 0\}$

Use this scenario for questions 1 through 6. Connie sells hot chocolate at a stand in a park. She records the number of cups of hot chocolate she sells, as well as the average daily low temperature. The table shown indicates the data for 12 days in January.

Hot Chocolate Sales

Average Temperature (°F)	59	63	55	50	49	40	45	43	47	44	50	48
Cups of Hot Chocolate Sold	60	55	70	82	85	102	95	90	93	75	78	91

Connie used technology to generate a scatter plot and a line of best fit, as shown.



- Write the equation for the line of best fit using the values generated with technology.
 $y = -1.91278x + 175.857$
- Interpret the slope of the line of best fit in the context.
There are 1.91278 less cups of hot chocolate sold per degree increase of temperature.
- Describe the strength of the association between temperature and the number of cups of hot chocolate Connie sells.
Since the r (correlation coefficient) is near -1 , this shows there is a strong association between temperature and the number of cups of hot chocolate sold.

4. Describe the direction of the linear association between temperature and the number of cups of hot chocolate Connie sells.

As the temperature increases, the number of cups of hot chocolate decreases – resulting in a negative linear association.

5. Suppose Connie knows that tomorrow's temperature is predicted to be 40°F. Based on the line of best fit, how many cups of hot chocolate should she anticipate selling?

$$y = -1.91278(40) + 175.857$$

$$y = 99.3458$$

99 cups

6. A few weeks ago, Connie sold 120 cups of hot chocolate. Based on the line of best fit, what was the average temperature that day?

$$120 = -1.91278x + 175.857$$

$$-120 = -1.91278x - 175.857$$

$$\frac{-55.857}{-1.91278} = \frac{-1.91278x}{-1.91278}$$

$$x = 29.2$$

29°F

A scatter plot created with technology generated the following values, as shown. Use the values for questions 7 through 10.

$$y_1 \sim mx_1 + b$$

STATISTICS

$$r^2 = 0.8913$$

$$r = 0.9441$$

PARAMETERS

$$m = 0.860687$$

RESIDUALS

$$e_1 \text{ plot}$$

$$b = 0.645038$$

7. Describe the strength of the linear association for this data.
Since the r (correlation coefficient) is close to 1, there is a strong association.
8. Describe the direction of the linear association of this data.
Because the slope (or the correlation coefficient) is positive, the direction of the line of best fit is positive – increasing from left to right.
9. Use this data to write the equation of the line of best fit in slope-intercept form.
 $y = 0.860687x + 0.645038$
10. Use the line of best fit to predict a value for y if the value of x is 18.
 $y = 0.860687(18) + 0.645038$
 $y = 16.137404$

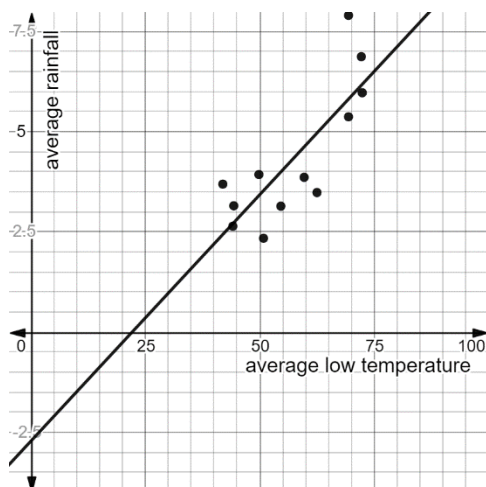
Algebra 1
Unit 3
Lesson 7 Homework

Name Answer key
Date _____
Period _____

A table shows the average low temperature and the average rainfall each month for a year in Jacksonville, Florida.

Month	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec
Average Low Temperature (°F)	41.9	44.3	49.8	54.6	62.5	69.4	72.4	72.2	69.4	59.7	50.8	44.1
Average Rainfall (in)	3.69	3.15	3.93	3.14	3.48	5.37	5.97	6.87	7.9	3.86	2.34	2.64

The scatter plot shown represents the data from the table with a linear equation fit to model the equation.



This screenshot shows the slope, y -intercept, and correlation coefficient associated with the linear relationship modeled on the scatter plot.

$$y_1 \sim mx_1 + b$$

STATISTICS

$$r^2 = 0.6461$$

$$r = 0.8038$$

PARAMETERS

$$m = 0.122798 \quad b = -2.71046$$

RESIDUALS

$$e_1 \text{ plot}$$

- Write the equation for the line of best fit shown.

$$y = 0.122798x - 2.71046$$

2. Interpret the meaning of the slope in terms of the context.

As the average low temperature increases by 1° , the average rainfall increases by 0.122798 in.

3. Interpret the meaning of the y -intercept in terms of the context.

When the average low temperature is 0°F , the rainfall is -2.71046 in. which is not possible.

4. Interpret the meaning of the correlation coefficient in terms of the context.

Strong, Positive, Linear

5. Use an inequality to express an appropriate domain in terms of the context.

Answers may vary.
 $35 \leq x \leq 75$

6. Use an inequality to express an appropriate range in terms of the context.

Answers may vary.
 $0 \leq y \leq 8$

7. Use the model to predict the average rainfall in a month with an average low temperature of 74°F .

$$y = 0.122798(74) - 2.71046$$
$$y \approx 6.4"$$

The average rainfall predicted would be 6.4in.

8. Use the model to predict the average rainfall in a month with an average low temperature of 53°F .

$$y = 0.122798(53) - 2.71046$$
$$y \approx 3.8\text{in.}$$

The predicted average rainfall would be 3.8 in.

9. Use the model to predict the average temperature in a month with an average rainfall of 2 inches.

$$2 = 0.122798x - 2.71046$$
$$x \approx 38.4^\circ$$

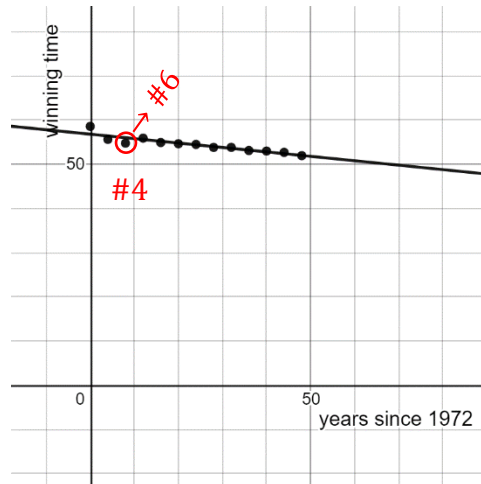
The predicted average temperature would be 38.4°

10. Use the model to predict the average temperature in a month with an average rainfall of 6 inches.

$$6 = 0.122798x - 2.71046$$
$$x \approx 70.9^\circ$$

The predicted average temperature would be 70.9°

A scatter plot representing the gold medal winning times for the womens' 100-meter freestyle swimming race since 1972 is shown.



The raw data used to construct the scatter plot is shown.

Year	1972	1976	1980	1984	1988	1992	1996
Winning Time (in seconds)	58.59	55.65	54.79	55.92	54.93	54.65	54.5
Year	2000	2004	2008	2012	2016	2020	
Winning Time (in seconds)	53.83	53.84	53.12	53	52.7	51.96	

The equation of the line of best fit that models this data is $f(x) = -0.1x + 56.83$, where f is the winning time of the womens' 100 meter freestyle even, and x is the number of years since 1972.

- How many years after 1972 was the 2016 Olympics?

44 years

- Using the equation of the line of best fit, find the predicted finishing time of the gold medalist in 2016.

$$f(x) = -0.1(44) + 56.83 \quad f(x) = 52.43$$

The predicted finishing time is 52.43 seconds.

- Identify the residual value of the point representing the 2016 winning time.

$$52.7 - 52.43 = 0.27$$

4. Identify one point on the scatter plot where the predicted value is above the actual value.

1980 Predicted: 56.03
Actual : 54.79

5. Find the residual value of the point selected for question 4.

$$54.79 - 56.03 = -1.24$$

6. Identify one point on the scatter plot where the predicted value is below the actual value.

1984 Predicted: 55.63
Actual : 55.92

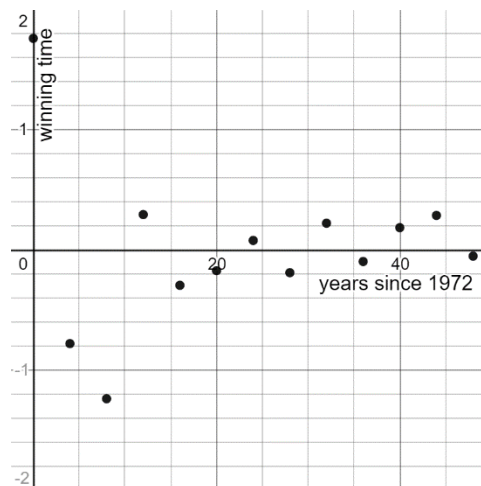
7. Find the residual value of the point selected for question 6.

$$55.92 - 55.63 = 0.29$$

8. Does the data appear to have a strong linear correlation? Explain how you know.

Because the data fits so closely with the line of fit, it is a strong linear correlation.

The residual plot for the scatter plot is shown.



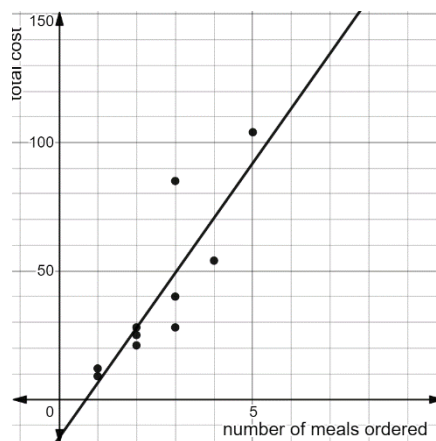
9. Explain how the residual values support the answer from question 8.

Because so many residuals are less than 1 & greater than -1 on the y-axis, this supports the idea that because the residuals are so small it is a stronger linear correlation.

A restaurant took a random sample of customers ordering takeout meals. The data from the sample is shown.

Number of Meals ordered	1	2	5	3	2
Total Cost	\$12	\$21	\$104	\$28	\$25
Number of Meals ordered	4	3	3	1	2
Total Cost	\$54	\$85	\$40	\$9	\$28

This data is represented by a scatter plot with the function $c(x) = 21.42x - 15.08$ fit to the data, where c represents the total cost of the order and x represents the number of meals ordered.



For questions 1 through 5, complete the tables to find the residual values for each number of meals ordered.

1.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
1	\$12	\$ 6.34	5.66
1	\$9	\$ 6.34	2.66

2.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
2	\$21	\$ 27.76	-6.76
2	\$25	\$ 27.76	-2.76
2	\$28	\$ 27.76	0.24

3.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
3	\$28	\$ 49.18	-21.18
3	\$40	\$ 49.18	-9.18
3	\$85	\$ 49.18	35.82

4.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
4	\$54	\$ 70.60	-16.6

5.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
5	\$104	\$ 92.02	11.98

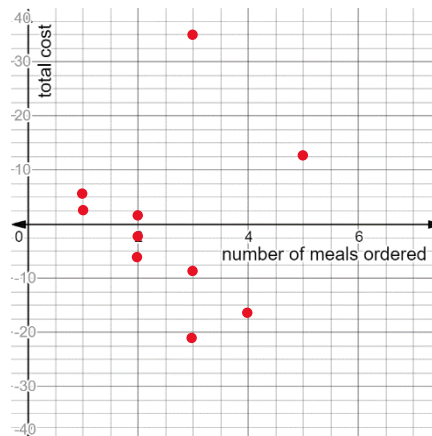
6. How many positive residuals does the data set have?

5

7. What is the residual with the largest absolute value?

35.82 (3 meals with actual cost \$85)

8. Plot the residuals on the coordinate grid.



9. Explain whether the linear function is a good fit for this data set. Answers may vary

The residual values are quite spread out & fairly large, so I do not think the linear function is a good fit for this data set.